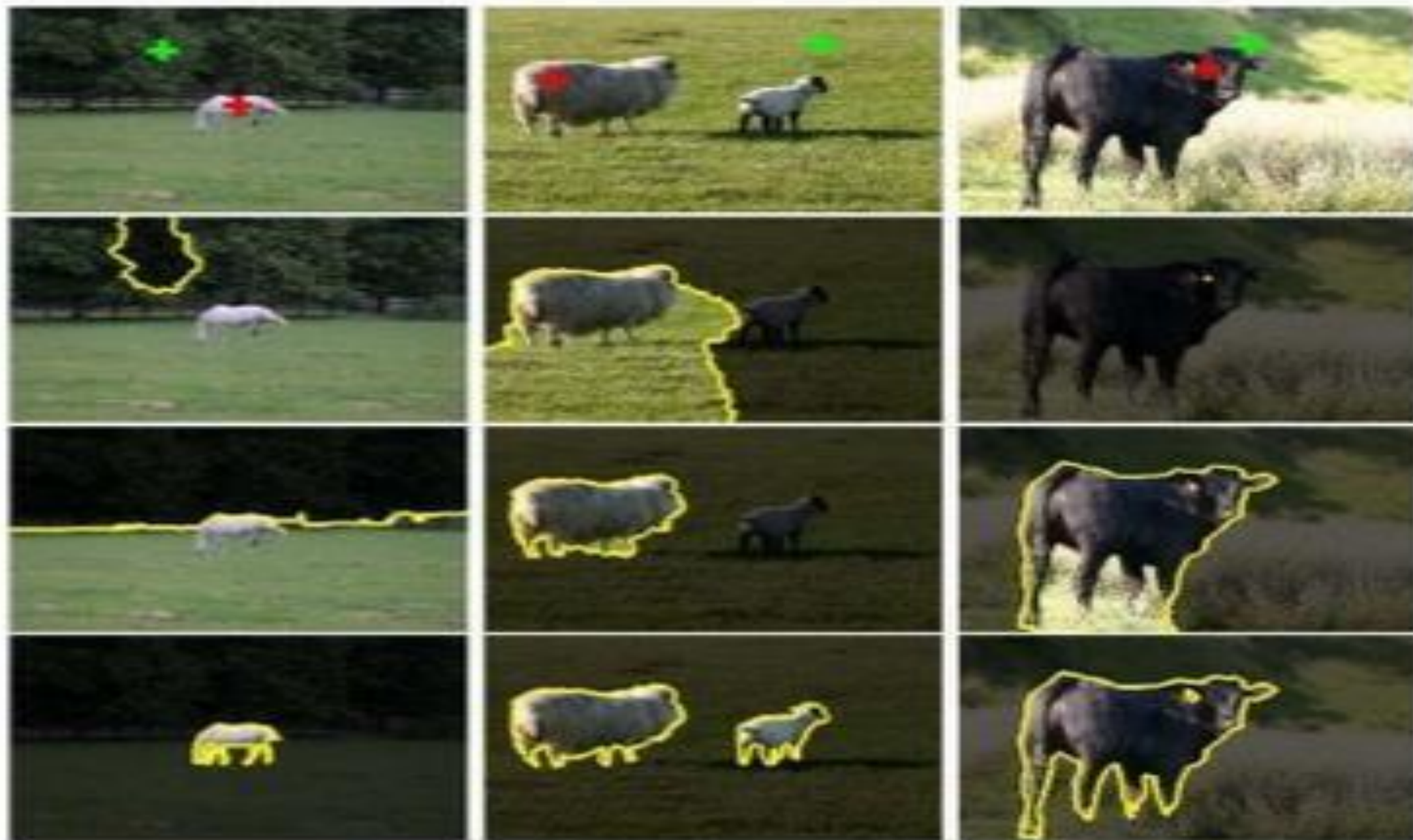


Image Segmentation

Unit-6

Introduction

- Image segmentation is a process of partitioning a digital image into multiple regions (set of pixels).
- The goal of segmentation is to simplify and change the representation of an image into something that is more meaningful and easier to analyze.
- It is typically used to locate objects and boundaries (line and curves) in an image.
- The result of an image segmentation is a set of regions that collectively cover the entire image.
- Each of the pixel in a region are similar with respect to the some characteristics such as color intensity or texture.
- Segmentation accuracy determines the success or failure of computerized analysis procedure.



Application

Some practical application of image segmentation are as follows:

1. Medical Imaging
 - Measurement Of Tissue Volumes
 - Diagnosis
 - Treatment Planning
 - Study Of Anatomical Structure
2. Locate Object In Satellite Image
3. Face Recognition
4. Finger Print Recognition
5. Machine Vision
6. Iris Recognition

Segmentation

- Segmentation accuracy determines the eventual success or failure of computerized analysis procedures.
- For this reason, considerable care should be taken to improve the probability of accurate segmentation.

What is Segmentation? (Cont...)

- Segmentation accuracy determines the eventual success or failure of computerized analysis procedures.
- For this reason, considerable care should be taken to improve the probability of accurate segmentation.



Algorithms

Image segmentation algorithm are generally based on two properties:

1. Discontinuity
2. Similarity (continuity)

Discontinuity

- This approach is to partition an image based on abrupt changes in intensity such as edges in an image.
- There are three basic types of gray level discontinuities in an image i.e. points or spots, lines and edges.

Continuity

- This approach is based on partitioning image into regions that are similar according to pre-defined criteria such as thresholding, region growing, region merging, and splitting.

Points, Lines and Edge Detection

- The focus of this topic is on segmentation methods that are based on detecting sharp, local changes in intensity.
- Three types of images features in which we are interested are;
 - **Edges:** edge pixels are pixels at which the intensity of an image function changes abruptly, and edges are sets of connected edge pixels.
 - **Lines:** line may be viewed as a edge segment in which the intensity of the background on either side of the line is either much higher or much lower than the intensity of the line pixels.
 - **Isolated point:** It can be viewed as a line whose length and width are equal to one pixel.

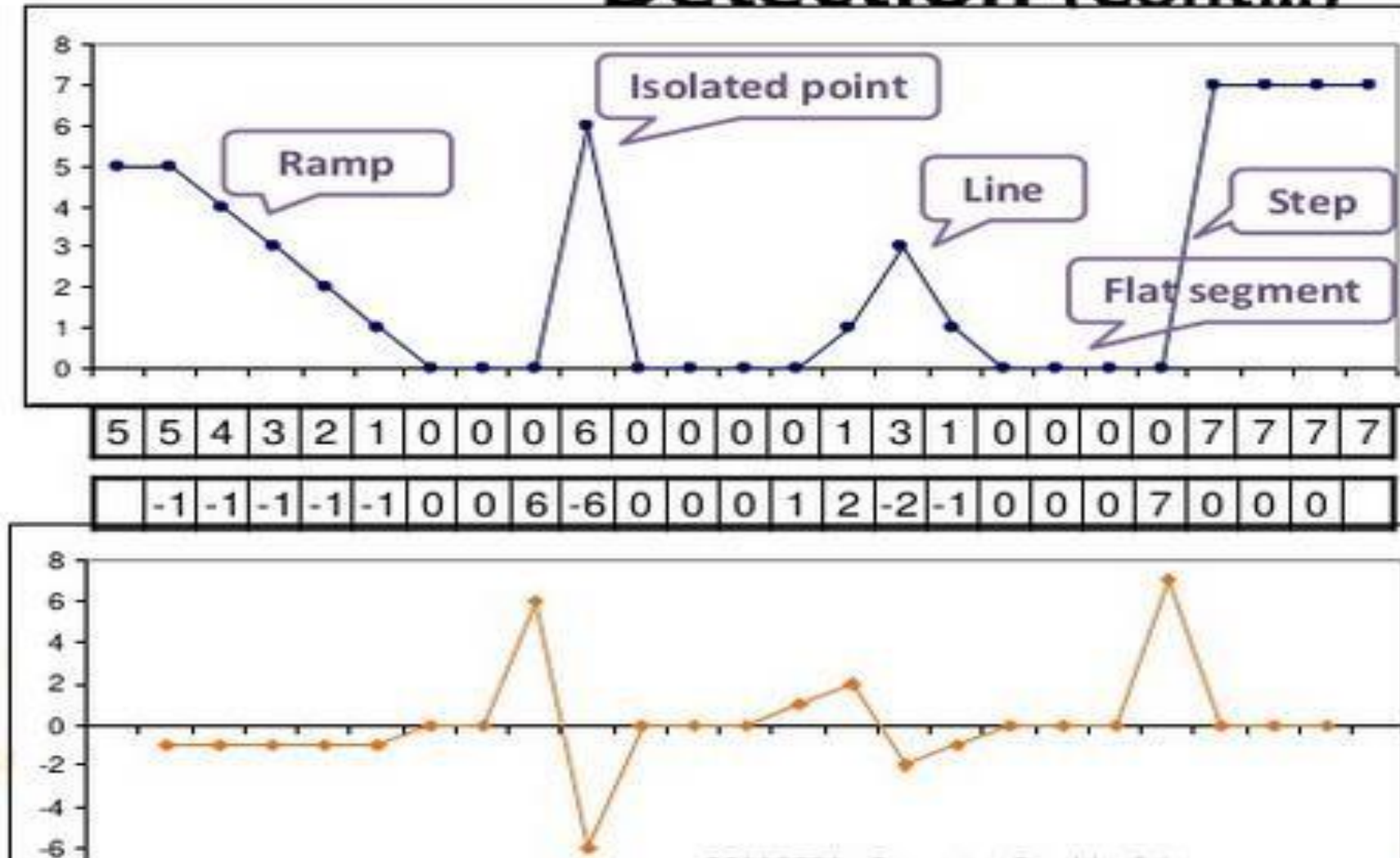


Background of Points, Lines and Edge Detection

- Local changes in intensity can be detected using derivatives.
- Derivatives of a digital function are defined in terms of differences.
- First order derivatives:
 - Must be zero in the areas of constant intensity.
 - Must be nonzero at the onset of an intensity step or ramp.
 - Must be nonzero along the ramp.



Background of Points, Lines and Edge Detection (Cont...)

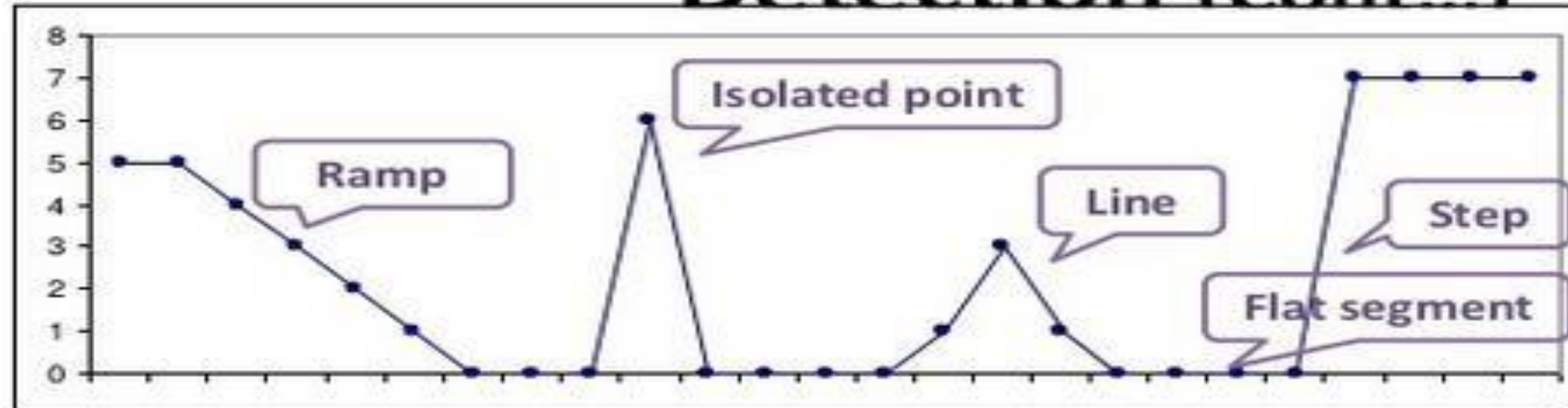


Background of Points, Lines and Edge Detection (Cont...)

- Second order derivatives:
 - Must be zero in the areas of constant intensity.
 - Must be nonzero at the onset and end of an intensity step or ramp.
 - Must be zero along the ramp of constant slope.
- First order derivatives provides thick edges and second order derivatives provides much finer ones (including noise).

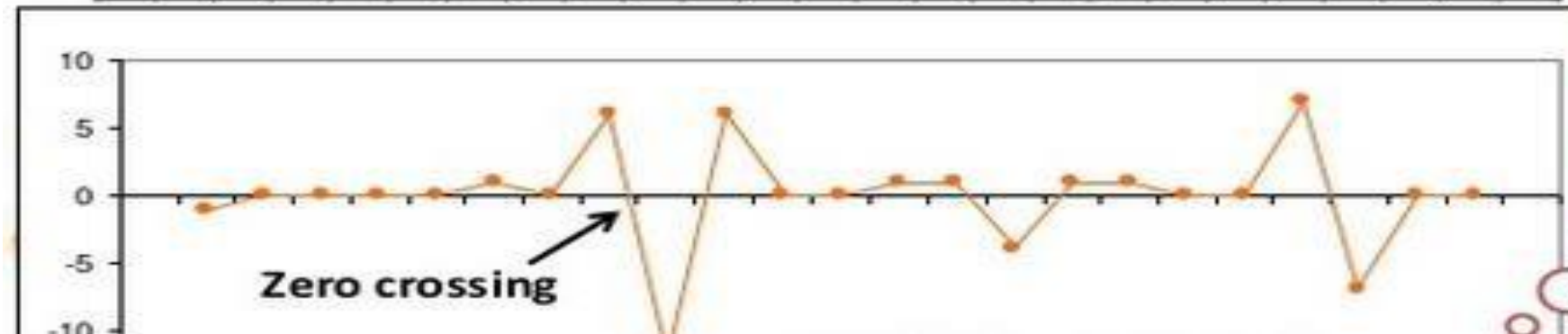


Background of Points, Lines and Edge Detection (Cont...)



5	5	4	3	2	1	0	0	0	6	0	0	0	0	1	3	1	0	0	0	0	7	7	7	7
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

	-1	0	0	0	0	1	0	6	-12	6	0	0	1	1	-4	1	1	0	0	7	-7	0	0
--	----	---	---	---	---	---	---	---	-----	---	---	---	---	---	----	---	---	---	---	---	----	---	---



Zero crossing property is quite useful for locating edges

Background of Points, Lines and Edge Detection (Cont...)

- **Double Edge Effect in Second Order Derivatives:**
 - In both the ramp and step edges that the second derivatives has opposite signs (-ve to +ve or vise-versa) as it transitions into and out of an edge.
 - This is known as double edge effect and it is important to locate edges.
- The sign of the second derivative is used also to determine whether an edge is a transition from light to dark (-ve $\rightarrow$ +ve second derivative) or from dark to light (+ve $\rightarrow$ -ve second derivative), where the sign is observed as we move into the edge.



Edge Models

- Edge detection is the approach used most frequently for segmenting images based on abrupt(local) changes in intensity.
- Edge models are classified according to their intensity profiles as follows:
 - Step edges
 - Ramp edges
 - Roof edges



Edge Models (Cont...)

- **Step Edges:**

- A step edge involves a transition between two intensity levels occurring ideally over the distance of 1 pixel.
- Canny edge detection algorithm was derived using a step edge model.



Edge Models (Cont...)

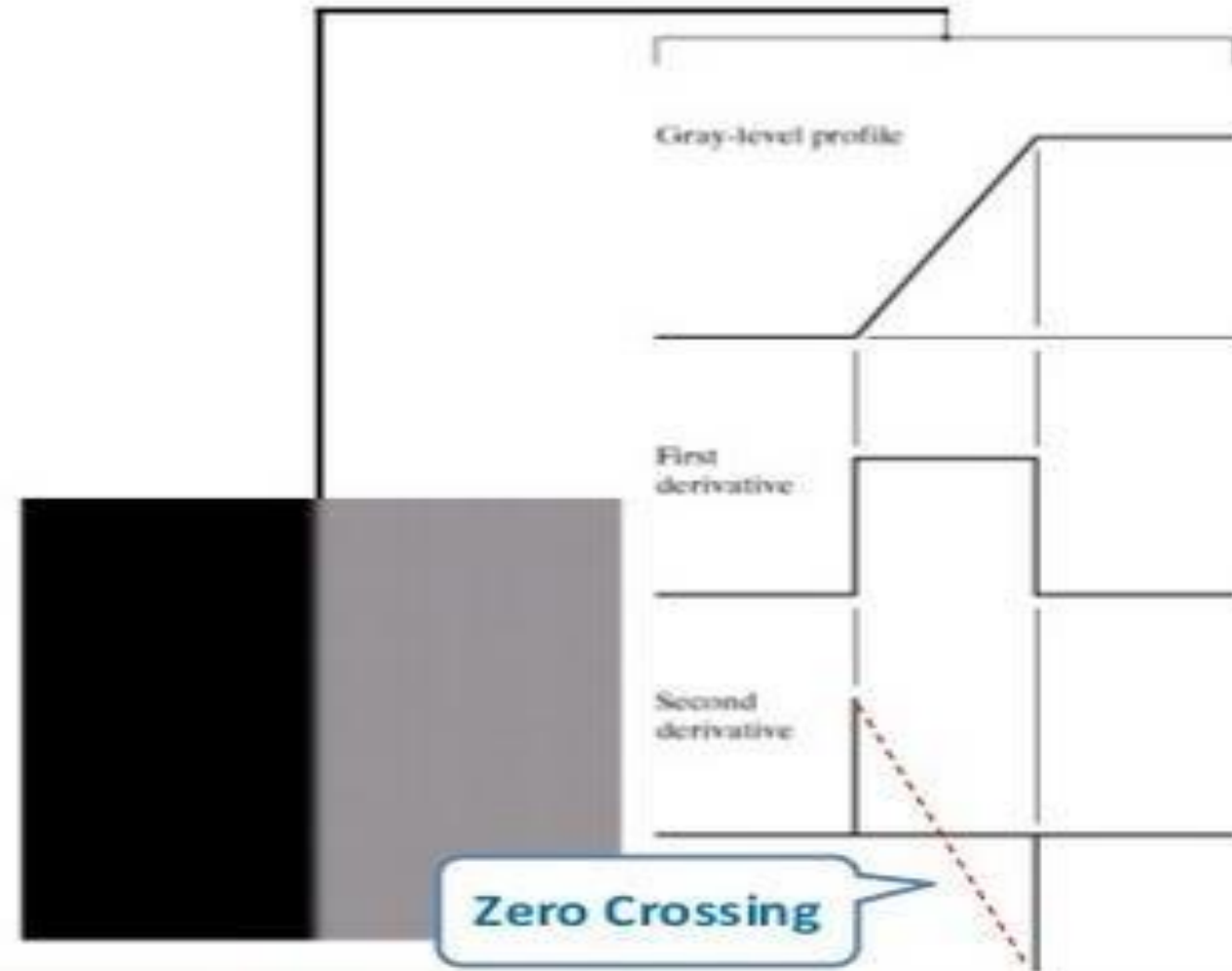
- **Ramp Edges:**

- In practice, digital images have edges that are blurred and noisy.
- Blurring determined by limitations in the focusing mechanism and the noise level determined principally by the electronic components of the imaging system.
- In such situations, edges are more closely modeled as having an intensity ramp profile.
- The slope of the ramp is inversely proportional to the degree of blurring in the edge.
- It does not contains a thin path.



Edge Models (Cont...)

- 1st derivative tells us where an edge is
- 2nd derivative can be used to show edge direction
- Zero crossings can be used for locating the canters of thick edges.

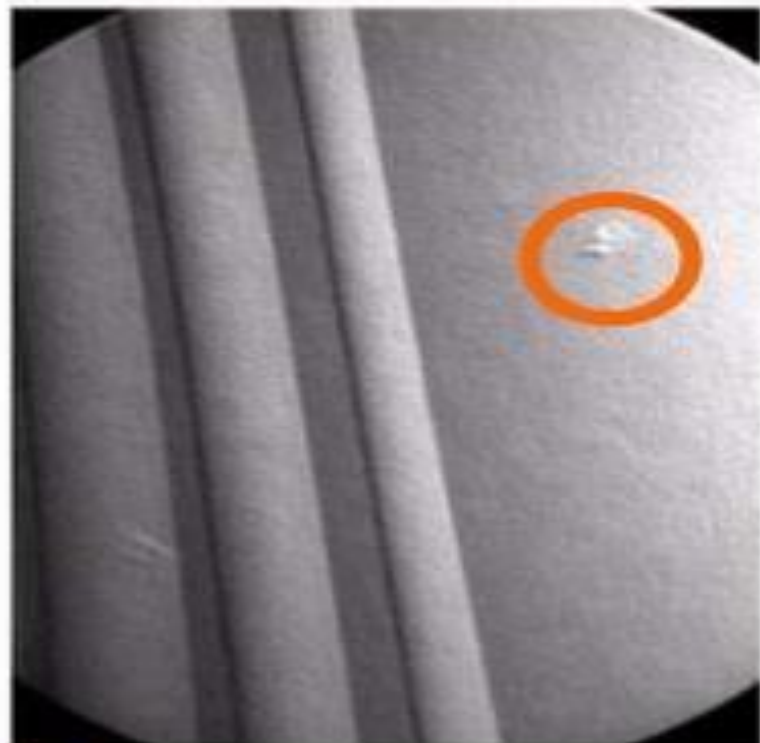


Detection of Isolated Points

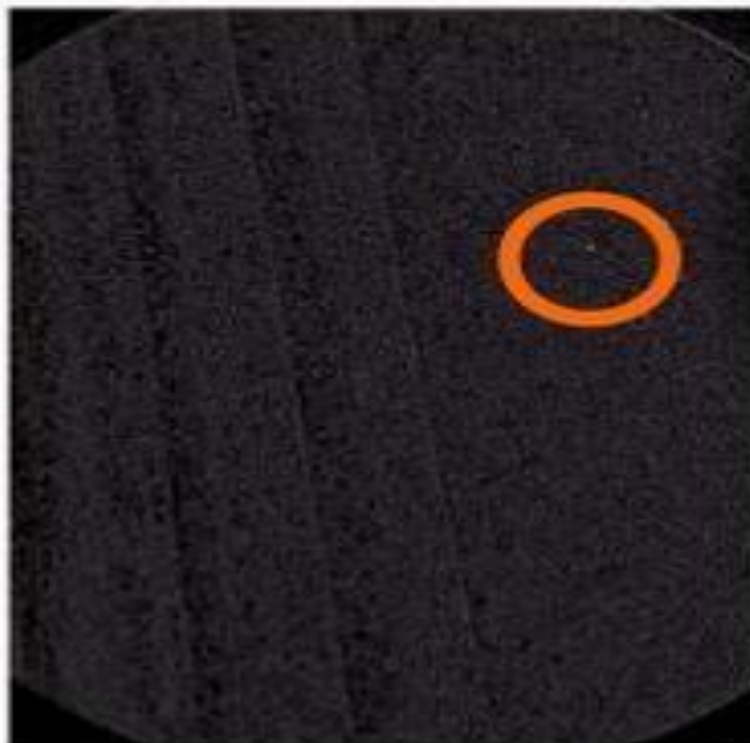
- Point detection is based on second derivatives.
- Point detection can be achieved simply using the Laplacian mask given below:

-1	-1	-1
-1	8	-1
-1	-1	-1
- Points are detected at those pixels in the subsequent filtered image that are above a set threshold

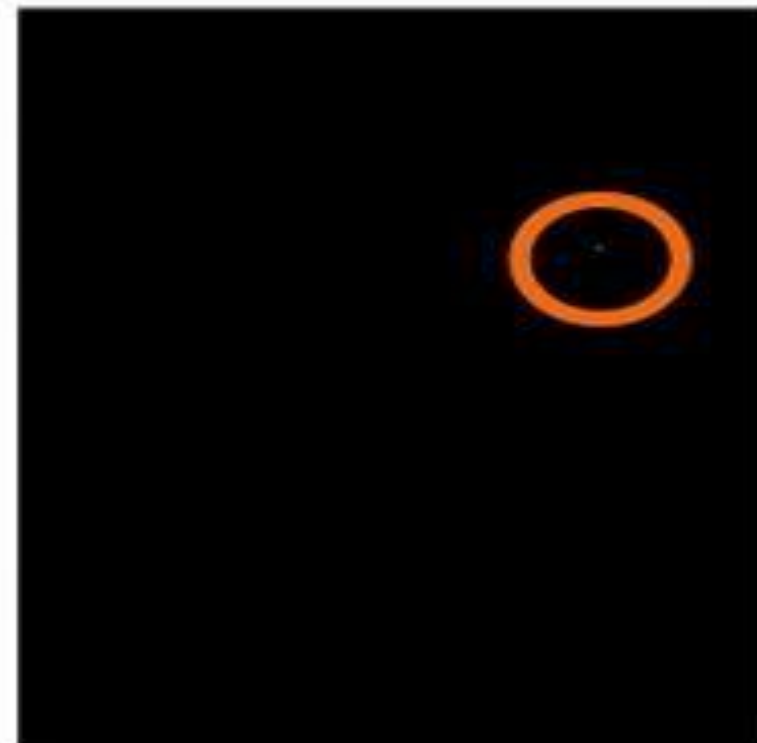
Detection of Isolated Points (Cont...)



X-ray image of a
turbine blade



Result of point
detection



Result of
thresholding

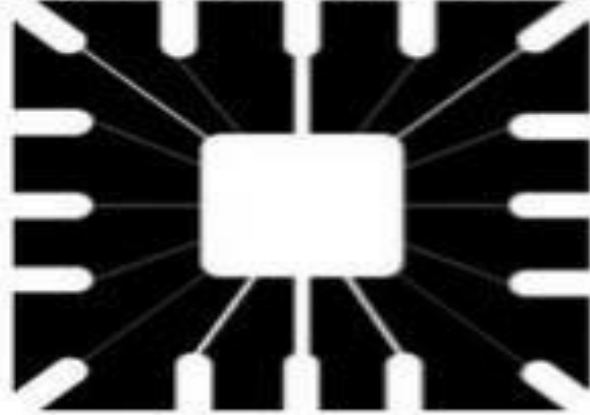
Line Detection

- We can use Laplacian mask for line detection.
- It is assumed that the lines are thin with respect to the size of the detector.
- The masks below will extract lines that are one pixel thick and running in a particular direction.

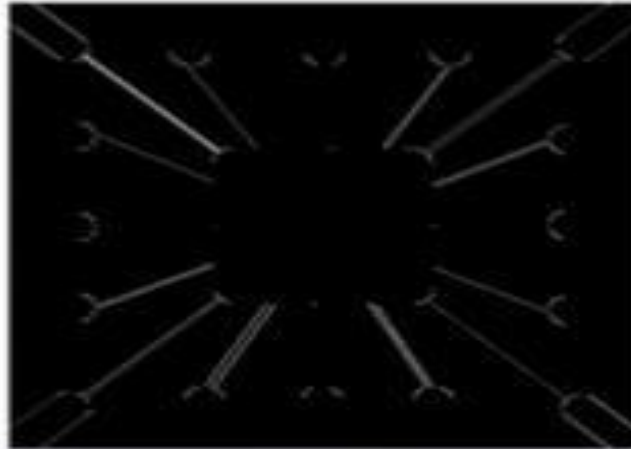
-1	-1	-1	-1	-1	2	-1	2	-1	2	-1	-1
2	2	2	-1	2	-1	-1	2	-1	-1	2	-1
-1	-1	-1	2	-1	-1	-1	2	-1	-1	-1	2
Horizontal			+45°			Vertical			-45°		

Line Detection (Cont...)

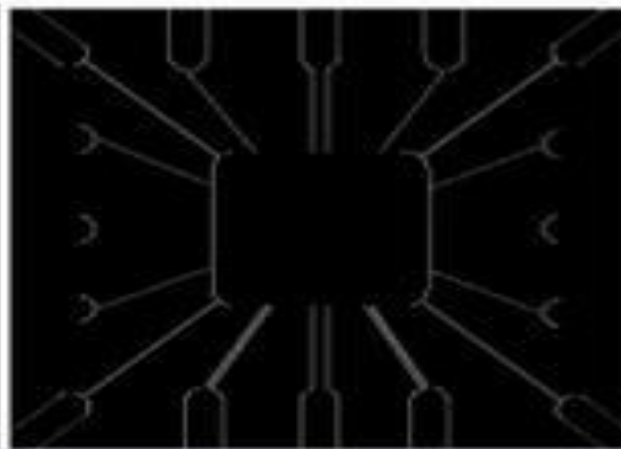
Binary image of a wire bond mask



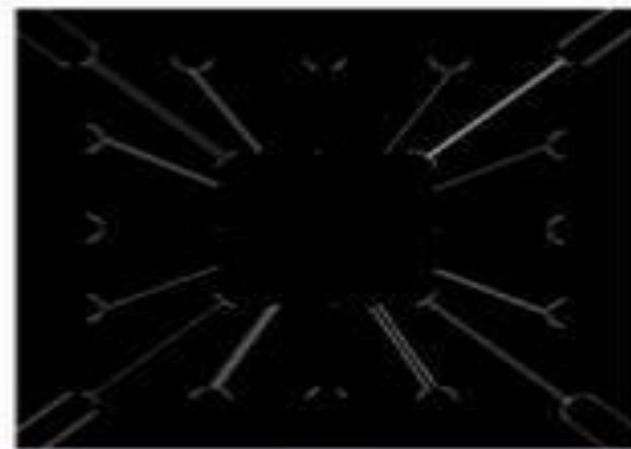
Horizontal



+45°



Vertical



-45°

Edge Detection with Image Gradient

- Given a 3*3 region of an image the following edge detection filters can be used under the 1st order derivatives.

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

-1	0	0	-1
0	1	1	0

Roberts

-1	-1	-1	-1	0	1
0	0	0	-1	0	1
1	1	1	-1	0	1

Prewitt

-1	-2	-1	-1	0	1
0	0	0	-2	0	2
1	2	1	-1	0	1

Sobel

Edge Linking & Boundary Detection:

- Ideally edge detection should yield sets of pixels only on edges.
- Actually, these pixels seldom characterize edges completely because of noise, breaks in edges due to non uniform illumination, & other effects that introduce spurious discontinuities in intensity values.
- So, Edge detection is followed by edge linking algorithm.

Local Processing

- One of the simplest approach for edge linking is to analyze the characteristics of pixels in a small neighborhood about every point (x, y) that has been declared an edge point.
- Two principal properties used for establishing similarity of edge pixels in this kind of analysis are:
 - 1) The strength (magnitude)
 - 2) the direction of the gradient vector.
- Let S_{xy} denote the set of coordinates of a neighborhood centered at point (x, y) in an image.
- An edge pixel with coordinates (s, t) in S_{xy} is similar in magnitude to the pixel at (x, y) if

$$|M(s, t) - M(x, y)| \leq E$$

- Where E is a positive threshold

- The direction angle of the gradient vector is given below.
- An edge pixel with coordinates (s, t) in S_{xy} has an angle similar to the pixels at (x, y) if

$$|\alpha(s, t) - \alpha(x, y)| \leq A$$

Where A is a positive angle threshold

- A pixel with coordinates (s, t) in S_{xy} is linked to the pixel at (x, y) if both magnitude & direction criteria are satisfied.

Global Processing using the Hough transform

- Here, we develop an approach based on whether sets of pixels lie on curves of a specified shape. Once detected, these curves form the edges or region boundaries of interest.
- Given n points in an image. Suppose that we want to find subset of those points that lie on straight lines.
- One solution is to find all lines determined by every pair of points & then find all subsets of points that are close to particular lines.
- This approach involves finding $n(n-1)/2 \approx n^2$ lines & then performing $n(n(n-1))/2 \approx n^3$ comparisons of every point to all lines.

Segmentation Techniques based on similarity

Types of image segmentation on the property of similarity:

1. Region Based Segmentation
2. Segmentation Using Region Growing
3. Using Region Splitting And Merging
4. Segmentation By Thresholding
5. Basic Global Thresholding

1. Region Based Segmentation

- The objective of image segmentation is to partitioned on image into multiple regions.
- This technique is based on finding the regions directly. To apply this technique following basic formulation condition must be applied:

Let R represent the entire image region, segmentation partition R into n -sub regions as $R_1, R_2, R_3, \dots, R_n$ such that:

1. $\bigcup_{i=1}^n R_i = R$
 - It indicates that every pixel must be in a region.
2. $R_i \cap R_j = \emptyset$
 - It indicates that the multiple regions must be disjoint.
3. $P(R_i) = \text{TRUE}$
 - It indicates that properties must be satisfied by all the pixels in a segmented regions i.e. all pixels in R_i have the same intensity level.
4. $P(R_i \cup R_j) = \text{False}$ for any adjacent regions R_i and R_j .
 - It indicates that properties of region R_i and R_j . are different.

2. Segmentation Using Region Growing

- Region growing is a technique that groups pixels or sub-regions into larger region based on pre-defined criteria.
- Let us pick up an arbiter pixel (r, c) from the domain of an image to be segmented. This pixel is called **seed pixel**.
- The basis approach is to start with a set of seed point. Then, examine the nearest neighbor (i.e. 4 or 8 neighbor) of (r, c) one by one and neighborhood pixel accepted belongs to the same region as (r, c) .
- Once, a new pixel is accepted due to the homogeneity property of a region as a member of the current region. Then, the nearest neighbor of this new pixel are examined.

2. Segmentation Using Region Growing

- The key factors in region growing are as follows:
 1. The choice of Similarity Criteria
 2. The selection of seed points
 3. The definition of a stopping rule

Limitation of Region Growing

- It is not very stable: significantly different results are obtained when switching between 4-connectivity and 8-connectivity criteria.
- Segmentation results are very sensitive to choice of logical uniformity predicate.
- The number of seeds provided by the user may not be sufficient to assign every pixel to a region.
- If two or more seeds that should belong to the same region are incorrectly provided to the algorithm, it will be forced to create distinct regions around them although only one region should exist.

3. Using Region Splitting And Merging

- Region splitting is a top-down approach to image segmentation. It starts from the entire image and partitions it into smaller sub images until each resulting region is considered homogeneous by some criteria.
- At the end of the process, it is guaranteed that the resulting regions satisfy the homogeneity criterion.
- If two or more adjacent regions are similar enough that should be combined into one. This is the goal of merging step.
- The data structure most commonly used for this algorithm is the quadtree, a special type of tree in which each node has four children. Each leaf node in the quadtree corresponds to a region in the segmented image.

3. Using Region Splitting And Merging

Algorithmic form:

1. Define a logical uniformity predicate $P(R_i)$.
2. Compute $P(R_i)$ for each region.
3. Split into four disjoint quadrants any region R_i for which $P(R_i) = \text{FALSE}$.
4. Repeat steps 2 and 3 until all resulting regions satisfy the uniformity criterion, that is $P(R_i) = \text{TRUE}$.
5. Merge any adjacent regions R_j and R_k for which $P(R_i \cup R_k) = \text{TRUE}$.
6. Repeat step 5. until no further merging is possible.

3. Using Region Splitting And Merging

- Examples:

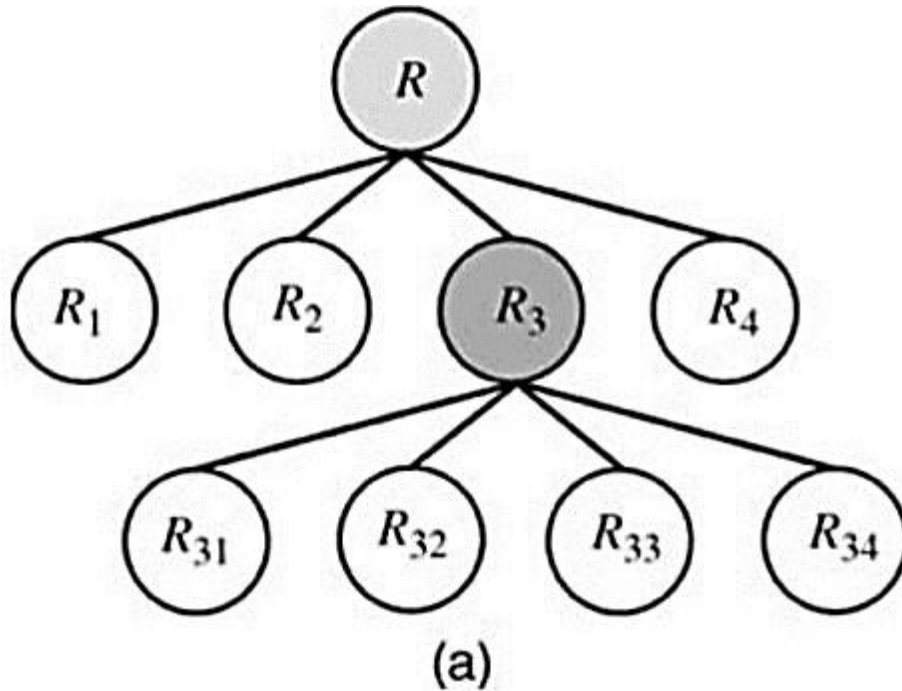


Figure: Corresponding Quad-tree

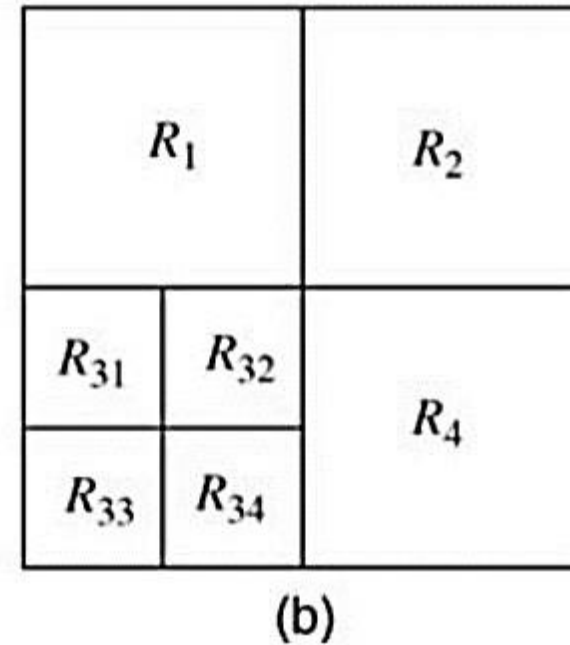


Figure: Partitioned Image

4. Segmentation by Thresholding

Foundation

- Suppose that the intensity histogram of an image $f(x, y)$ is composed of light object on a dark background as shown in figure below.

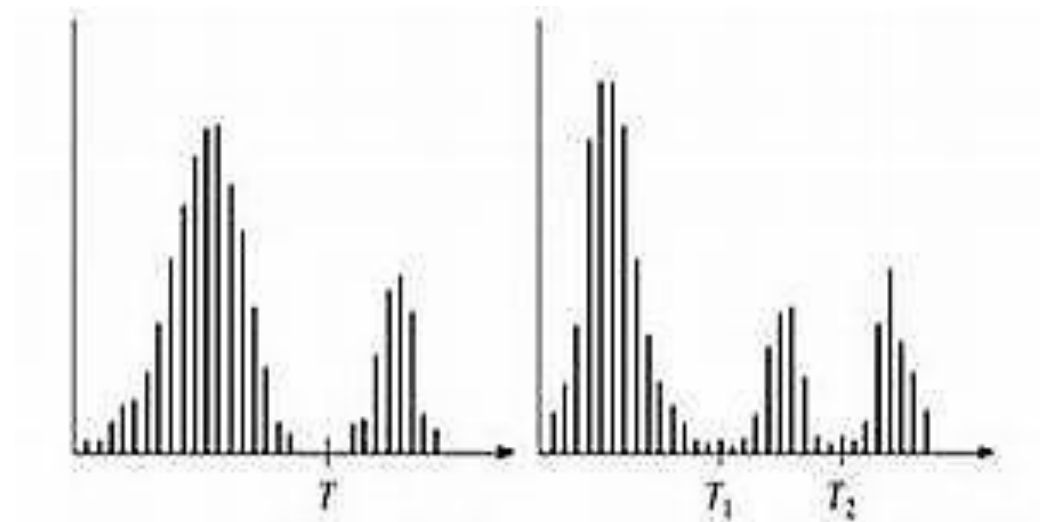


Figure: Intensity Histogram, (a) A single Threshold ; (b) Multiple Threshold

4. Segmentation by Thresholding (contd...)

- The figure (a) shows to extract the object from the background by selecting threshold T that separate the object from background. Then any point (x, y) for which $f(x, y) > T$ is called an *object point*; otherwise, the point is called *background point*.

- In other words, the segmented image, $g(x, y)$, is given by

$$g(x, y) = \begin{cases} 1 & \text{if } f(x, y) > T \\ 0 & \text{if } f(x, y) \leq T \end{cases}$$

- When T is a constant over an entire image, the process given in this equation is referred to as *global thresholding*.
- When the value of T changes over an image, we use the term *variable thresholding*.
- If T depends on the spatial coordinates (x, y) themselves, then variable thresholding is often referred to as dynamic or *adaptive thresholding*.

4. Segmentation by Thresholding (contd...)

- Figure (b) shows a slightly more difficult thresholding problem involving a histogram with three dominant modes. In this case, there are two types of light objects on the dark background.
- Here, *multi-level thresholding* classifies a point (x, y) as belonging to the background if $f(x, y) \leq T_1$, to one object class if $T_1 < f(x, y) \leq T_2$, and to other object class if $f(x, y) > T_2$.
- That is segmented image is given by

$$g(x, y) = \begin{cases} a & \text{if } f(x, y) \leq T_1 \\ b & \text{if } T_1 < f(x, y) \leq T_2 \\ c & \text{if } f(x, y) > T_2 \end{cases}$$

Where a, b, and c are any three distinct intensity values.

- Threshold can be viewed as operation that involves test against a function T of the form,

$$g(x, y) = T[(x, y), p(x, y), f(x, y)]$$

where,

$p(x, y)$ = some local property of the point

$f(x, y)$ = gray level of point (x, y)

- A threshold image can be defined as,

$$g(x, y) = \begin{cases} 1 & \text{if } f(x, y) > T \\ 0 & \text{if } f(x, y) \leq T \end{cases}$$

Thus, if T depends on both $f(x, y)$ and $p(x, y)$, the threshold is called as local and if T depends on spatial co-ordinate then, it is called Dynamic or Adaptive.

Factors affecting thresholding

- The success of intensity thresholding is directly related to the width and depth of the valley(s) separating the histogram modes. In turn, the key factors affecting the properties of the valley(s) are:
 1. The separation between peaks (the further apart the peaks are, the better the chances of separating the modes);
 2. The noise content in image (the modes broadens as noise increases);
 3. The relative sizes of objects and background;
 4. The uniformity of the illumination source; and
 5. The uniformity of reflectance properties of the image.

Basic Global Thresholding

- When the intensity distribution of objects and background pixels are sufficiently distinct, it is possible to use a single (global) threshold applicable over the entire image.
- So, in this approach, first partition the image histogram by using a single global threshold, ' T '.
- Segmentation is then accomplished by scanning the image pixel by pixel and labeling each pixel as object or background depending whether the gray level of the pixel is greater or less than the value of T .
- Following iterative algorithm can be used to obtain T :
 1. Select an initial estimate of T
 2. Segment the image using T . This will provide two group of pixels i.e. $G_1 > T$ and $G_2 \leq T$.
 3. Compute the average(mean) gray level intensity values μ_1 and μ_2 for the pixel in the regions G_1 and G_2 .
 4. Compute a new threshold value: $T = \frac{1}{2}(\mu_1 + \mu_2)$
 5. Repeat 2 through 4 until the difference between values of T in successive iterations is smaller than a predefined parameter ΔT .

Template Matching

- Template Matching is a technique in DIP for finding small parts of an image that match a template image.
- It can be used in manufacturing as a part of quality control as a way to detect edges in a image.
- Here, we have a template $g(i, j)$ and we wish to detect its instances in an image.
- The incoming is compared directly to copies (templates) stored in the memory. So it has various different applications such as: face recognition, finger print detection and medical image system.
- In this technique, the behind logic is that comparing between pixels or patterns. This method is also called *linear spatial filtering*.
- This is done only through the use of correlation Theorem.

Correlation Theorem

- It is the measure of degree to which two variables agree or not in a actual value.
- Here, two variables are the corresponding pixel values in two images i.e. template and source image. So, it is used to detect location of a certain object inside an image.
- Therefore, correlation coefficient is defined as:

$$\text{Correlation} = \frac{\sum_{i=0}^{N-1} (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=0}^{N-1} (x_i - \bar{x})^2 \sum_{i=0}^{N-1} (y_i - \bar{y})^2}}$$

where,

x_i = Template Gray Level Image

y_i = Source Gary Level Image

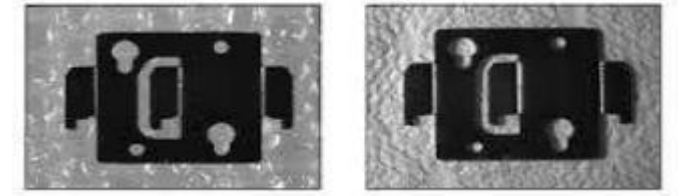
$\bar{x}$ = Average Gray Level Template Image

$\bar{y}$ = Average Gary Level Source Image

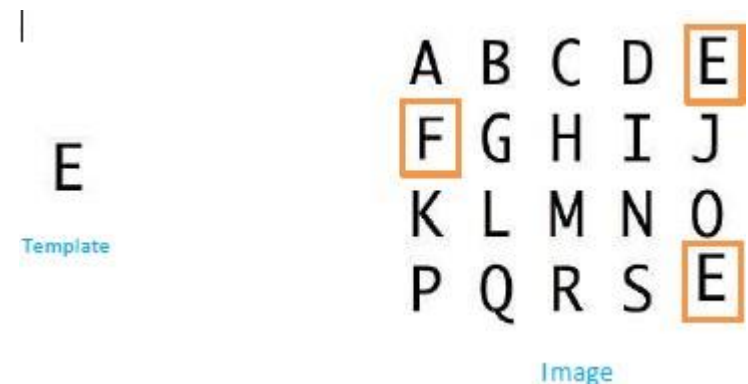
Template matching types

A. Pixel level Template matching

a) Total templates: Template is the *same size as the input image*. There is no rotation or translation invariance.

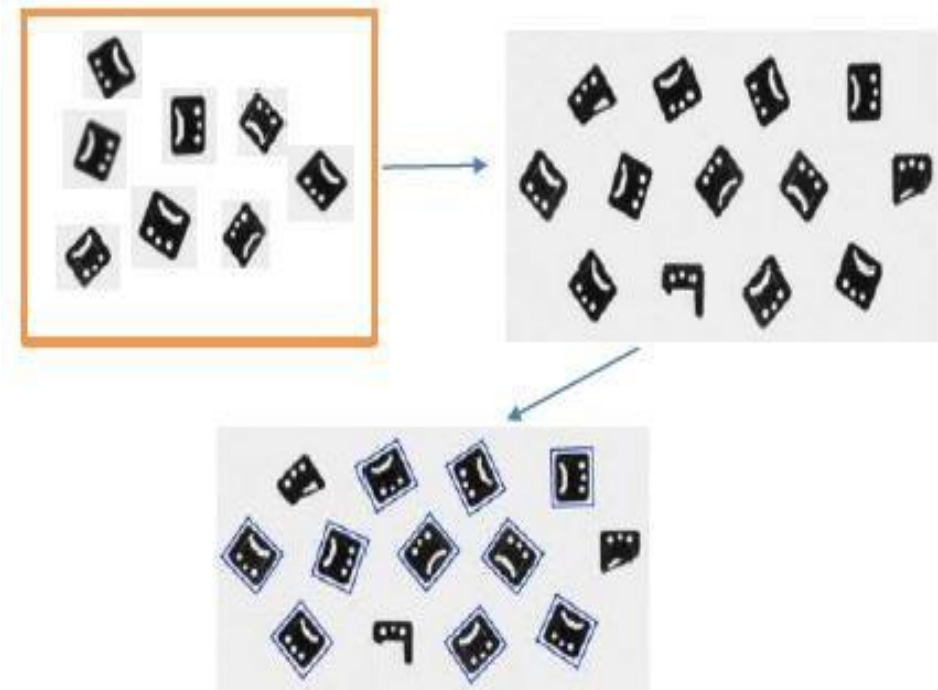
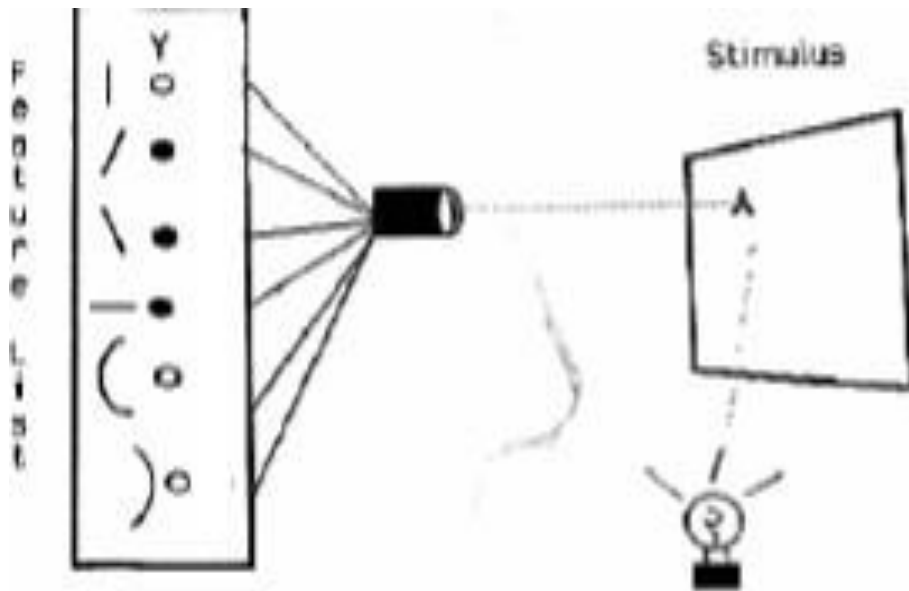


b) Partial templates: Template is *free* from the background. Multiple matches are allowed. Partial matches may also be allowed. Care must be taken in this case -- an **F** template could easily match to an **E**.



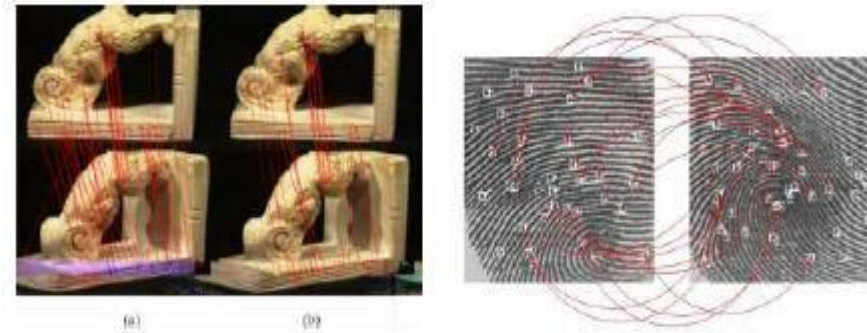
c) Piece templates : Templates that match one feature of a figure. These templates break a pattern into its component segments so, for example, "A" can be broken down into "/", "\", and "-". The order in which templates are compared to the scene is important: the largest templates must be tried first, since they contain the most information and may subsume smaller templates.

d)Flexible templates: These templates can handle stretching, misorientation and other possible deviations. A good prototype of a known object is first obtained and represented parametrically.



B. High level Template Matching

a) *Feature-based Matching:* When the template image has strong features, a feature-based approach may be considered; the approach may prove further useful if the match in the search image might be transformed in some fashion. Since this approach does not consider the entirety of the template image, it can be more computationally efficient when working with source images of larger resolution.



b) *Template-based Matching:* For templates without strong features, or for when the bulk of the template image constitutes the matching image, a template-based approach may be effective. Template-based template matching may potentially require **sampling** of a large number of points, it is possible to reduce the number of sampling points by reducing the resolution of the search and template images by the same factor and performing the operation on the resultant downsized images (multi-resolution, or pyramid, image processing).

Template based matching

- **Image Pyramid:** *Image Pyramid* is a series of images, each image being a result of down sampling (scaling down, by the factor of two in this case) of the previous element.



Level 0 (input image)



Level 1



Level 2

- **Pyramid Processing:** At each level of the pyramid, we will need appropriately downsampled picture of the reference template, i.e. both input image pyramid and template image pyramid (**Pyramid Processing**) should be computed.



Level 0 (template reference image)



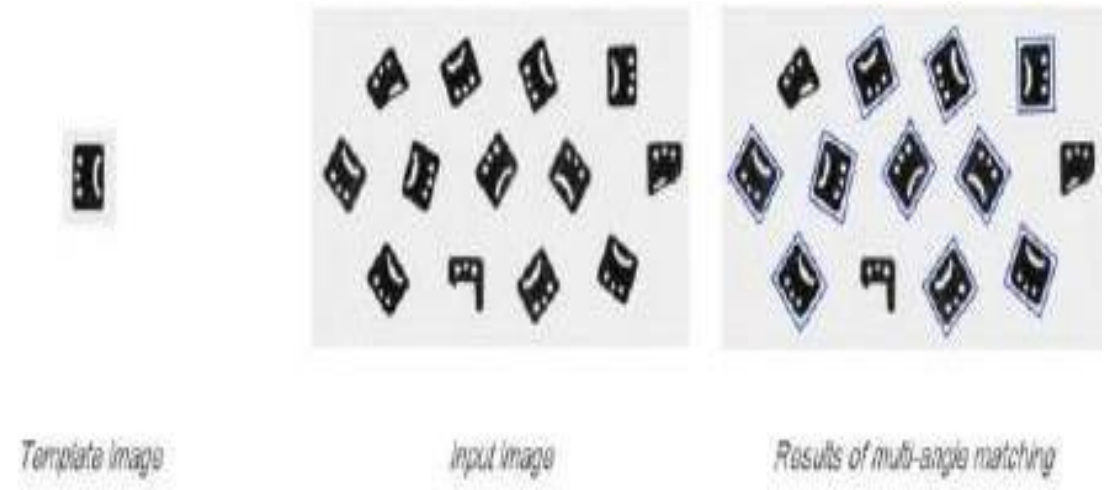
Level 1



Level 2

- **Grayscale-based Matching:** This is achieved by computing not just one *template image* pyramid, but a set of pyramids - one for each possible rotation of the template. During the pyramid search on the input image the algorithm identifies the pairs (***template position, template orientation***) rather than sole template positions. Similarly to the original schema, on each level of the search the algorithm verifies only those (*position, orientation*)pairs that scored well on the previous level (i.e. seemed to match the template in the image of lower resolution).

•The technique of pyramid matching together with *multi-angle* search constitute the **Grayscale-based Template Matching** method.



- **Edge-based Matching:** Edge-based Matching enhances the previously discussed Grayscale-based Matching using one crucial observation - that the shape of any object is defined mainly by the shape of its edges. Therefore, instead of matching of the whole template, we could extract its edges and match only the nearby pixels, thus avoiding some unnecessary computations. In common applications ***the achieved speed-up is usually significant.***

Grayscale-based Matching:



Edge-based Matching:

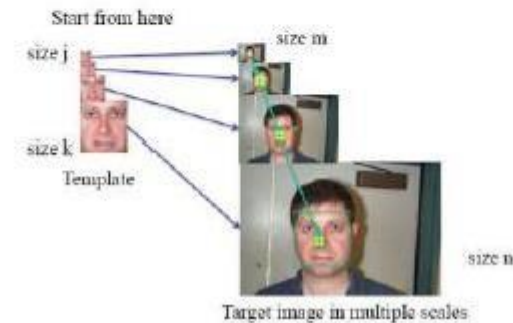


Template Matching Measures

- Measure of match between two images is considered to be a metric that indicate the degree of **similarity** or **dissimilarity** between them. Unless it is specifically stated otherwise, this metric can be increasing or decreasing with degree of similarity. Where the metric is specifically stated to be a measure of mismatch, it is a quantity that is increasing with the degree of dissimilarity.

Problems with template matching

- 1) The template represents the object as we expect to find it in the image
- 2) The object can indeed be scaled or rotated
- 3) This technique requires a separate template for each scale and orientation
- 4) Template matching become thus too expensive, especially for large templates
- 5) Sensitive to:
 - noise
 - occlusions



Template Matching Applications:

- 1) Template matching with various average face pyramid levels.
- 2) 3D reconstruction.
- 3) Motion detection.
- 4) Object recognition.
- 5) Panorama reconstruction.